Lesson 8.9 — Unit 8 Review and Assessment

Big idea: Unit 8 is statistical inference: starting from a sample, what can we say (and how confidently) about a whole population? Good design, honest margins of error, and disciplined hypothesis testing.

Quick review

Sampling: random samples; watch for bias.
Sampling distributions: SE = σ/√n for means, SE = √(p(1–p)/n) for proportions. CLT makes the sampling distribution approximately normal for large n.
Confidence intervals: point estimate ± z* · SE.
Hypothesis tests: H_0 vs. H_a; compute test statistic and p-value; reject if p-value < α.
Experiments: randomize, control, replicate, blind.

Worked example

Worked review. 90 of 200 sampled students prefer brand A. Construct a 95% CI for p.

p■ = 0.45. ME = 1.96 √(0.45 · 0.55/200) ≈ 1.96(0.0352) ≈ 0.069. CI: (0.381, 0.519).

Warm-ups

1. If μ = 50, σ = 10, n = 25, find SE(x■).
2. If p■ = 0.6 and n = 100, find SE(p■).
3. What z* gives a 95% confidence interval?

Practice

1. Construct a 95% CI for the proportion if 120 of 400 sampled support a measure.

2. A factory claim is 4% defective. A sample of 250 has 14 defects. Test H_0: p = 0.04 vs H_a: p > 0.04 at α = 0.05.

3. Identify the sampling method used: a teacher randomly chooses 5 students from each homeroom for a survey.

Challenge

1. A pollster wants ME = 0.025 (95% confidence) for a proportion. Using the worst case of p(1 – p), find the required sample size.

2. Test scores at School A have x■ = 78 with s = 10 in n = 50; School B has x■ = 75 with s = 12 in n = 60. Compute the SE of the difference (x■$_A$ – x■$_B$) under independence and decide whether the schools differ at the 5% level. (Use z = (3 – 0)/SE and a two-sided test.)

Answer key — Lesson 8.9

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. $10/\sqrt{25} = 2$.
2. $\sqrt{(0.6 \cdot 0.4/100)} = \sqrt{0.0024} \approx 0.049$.
3. 1.96.

Practice

1. p■ = 0.30. ME = $1.96\sqrt{(0.3 \cdot 0.7/400)} \approx 1.96(0.0229) \approx 0.045$. CI: (0.255, 0.345).
2. p■ = 14/250 = 0.056. $z = (0.056 - 0.04)/\sqrt{(0.04 \cdot 0.96/250)} = 0.016/0.01239 \approx 1.29$. p-value $\approx P(Z > 1.29) \approx 0.0985 > 0.05 \rightarrow$ fail to reject. Insufficient evidence the defect rate exceeds 4%.
3. Stratified sampling.

Challenge

1. $n \geq (1.96^2 \cdot 0.25)/(0.025)^2 = (3.8416)(0.25)/0.000625 \approx 1537$.
2. SE(x■$_A$ – x■$_B$) = $\sqrt{(s_A^2/n_A + s_B^2/n_B)} = \sqrt{(100/50 + 144/60)} = \sqrt{(2 + 2.4)} = \sqrt{4.4} \approx 2.10$. $z = 3/2.10 \approx 1.43$. Two-sided p-value $\approx 2(1 - 0.9236) \approx 0.153 > 0.05 \rightarrow$ fail to reject. Schools do not differ significantly at the 5% level.